Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**CS 535**
Program:	**MS(Data Science)**	**Semester:**	**Spring 2023**
Duration:	**30 Minutes**	**Total Marks:**	**13**
Paper Date:	**9-March-2023**	**Weight**	
Section:	**DS**	**Page(s):**	**4**
Exam:	**Quiz 2**		

Q1) Given the three-document corpus, answer the following question. [2+4 = 6 Marks]

d_1	information retrieval is process of index search retrieval
d_2	retrieval is used for evaluation of search results retrieval retrieval
d_3	evaluation in information in evaluation process search

(a) Convert document 2 in vector form using raw counts as weights. Clearly mention each dimension and its value.
(b) Convert the word "results" in form of vector. Context window size is 2. Use Tf.IDF weights.

Q2) Calculate following measures for the given confusion matrix.
Microaverage Precision
Microaverage Recall
Microaverage F-score
Macroaverage Precision
Macroaverage Recall
Macroaverage F-score

Gold Labels

		Coffee	Tea	Beans
System Output	Coffee	10	30	15
	Tea	40	5	25
	Beans	20	40	300